Hoang Hieu Long

Electrical Engineer | MSc Power Systems Engineering, University College London | London, UK
hoanghieulong200359@gmail.com | LinkedIn | longhoang-git.github.io

Profile

I am an electrical engineer with five years of experience across renewable energy consultancy and power-system research, specialising in HV cable systems, substation concepts and power-system studies. At OWC – an Aqualis Company, I have delivered electrical design for 20+ offshore wind, solar PV, BESS and interconnector projects from 50 MW to 2 GW, led three electrical work packages, one rated 9/10 by the client, and developed an inter-array cable layout optimiser that shortens cable length by 10–30%. Alongside my MSc in Power Systems Engineering at UCL, I continue to advise OWC on MV/HV design, technical due diligence and power-system studies.

Education

University College London (UCL), **United Kingdom** 2026–2027
MSc Power Systems Engineering

Hanoi University of Science and Technology (HUST), **Vietnam** 2020–2025
BSc Electrical Engineering; top 10% of the cohort; graduation thesis 10/10, ranked 1st of 200+ students

Shibaura Institute of Technology (SIT), **Japan** Apr–Nov 2023
Research exchange, Electromagnetic Actuator Laboratory

Professional and Research Experience

Industry

Consultant – Electrical Engineer, OWC – an Aqualis Company, remote 2026–Present

- Provide HV and MV analysis and advice on renewable energy projects in owner's engineer, technical due diligence and power-system study roles; maintain the team's analytical tools.

Electrical Engineer, OWC – an Aqualis Company, Vietnam 2024–2026

- **Scope:** designed and supported 20+ utility-scale projects (50 MW to 2 GW) in offshore wind, solar PV, BESS and HVAC interconnectors across the UK, Europe, MENA, APAC and the US, from feasibility to FEED.
- **Cable systems and substations:** delivered electrical design for 33 kV PV and BESS systems, 66–132 kV inter-array systems and export cable systems up to 275 kV, plus concept design of 230–500 kV offshore and onshore substations.
- **Power-system studies:** modelled reactive power compensation and losses for HVAC submarine cables over 100 km in DIgSILENT PowerFactory, for interconnector and offshore wind projects.
- **Techno-economics:** delivered CAPEX estimates, NPV-based design optioneering, technical due diligence and route-to-market risk assessments.
- **Leadership:** led 3 electrical work packages, coordinating 5+ multidisciplinary team members.
- **Engineering tools:** built 3 internal tools in Python, VBA and QGIS (IAC layout optimiser, microgrid failure assessment, Power-to-X emissions calculator), presented at WindEurope and used on 5+ live projects; quality-checked 5 in-house design tools used on 10+ projects a year.

Research

Hydrogen Team Leader, 100% Renewable Energy Laboratory, HUST 2022–2025
Supervisor: Associate Professor Nguyen Duc Tuyen

- Led a five-member team that published 2 journal articles and 3 conference papers on microgrid optimisation, energy management and PV–electrolyser direct coupling; won USD 5,000 of Green In funding for a green city model with hydrogen storage.
- **Energy management:** developed MATLAB models of an AC microgrid with battery–hydrogen storage and grid-forming control, including a state-space model for model predictive control (MPC) benchmarked against fuzzy-logic and state-machine strategies.
- **Hydrogen systems:** reviewed sizing and operating strategies for direct-coupled PV–electrolyser systems; developed a Power-to-Hydrogen life-cycle assessment method.

First Author, Book Chapter (accepted, in preparation), ASEAN Centre for Energy 2026

- Offshore wind financial model testing grid delay, curtailment and offtake risk in Viet Nam and the Philippines.

Selected Projects

1.32 GW offshore wind, Poland – *lead engineer, inter-array cable package* FEED
Led the team delivering the inter-array cable (IAC) package and liaised with subcontractors and the client; set turbines per IAC string from electrical losses and the cable schedule, and compared IAC designs with and without an offshore substation on NPV. Client feedback 9/10.

500–1,000 MW offshore wind, Morocco – *lead engineer, electrical work package* Feasibility
Led the electrical team and coordinated subcontractor and client inputs; designed IAC systems for four turbine-model and layout options at 66 kV and 132 kV, and compared them on electrical losses, cable schedule and NPV.

Up to 1,200 MW offshore wind, South Korea – *export system optioneering* Pre-FEED
Compared three export concepts for about 150 km of export cable on technical risk and cost-benefit: HVAC export via several offshore substations, HVAC export via a single offshore substation, and HVDC export.

400 MW Indonesia–Singapore HVAC interconnector – *main electrical designer* Interconnector
Designed the HVAC submarine cable system for the 100+ km route and modelled its reactive power compensation and transmission losses in DIgSILENT PowerFactory.

IAC layout optimiser – *developer and project manager, Python* OWC in-house tool
Took the tool from specification to release: a biased genetic algorithm (matrix operations, multiprocessing) that minimises IAC length under design constraints, producing a layout in 2–5 hours instead of days, with 10–30% shorter cable on past projects.

2 GW floating offshore wind, the Philippines – *route-to-market assessment* Market study
Assessed the supply–demand balance and transmission curtailment risk to test whether the project could feasibly connect to the grid, then turned the findings into an investment-risk table and alternative route-to-market options.

50 MW onshore wind farm, Vietnam – *lead electrical reviewer* Technical due diligence
Reviewed the grid-connection design and identified network overloads under normal and N-1 contingency conditions, with recommendations ahead of the investment decision.

Hydrogen strategy benchmarking for Vietnam – *author, 100% Renewable Energy Laboratory* Policy research
Authored 5 policy and market briefs that benchmarked national hydrogen strategies (US, Germany, Japan, China) on long-term targets, demand creation, investment de-risking and R&D support, and set out policy options for Vietnam, including emissions trading, tax incentives and the cost case for hydrogen co-firing in gas plants.

Publications

1. **Long HH**, Huong HL, Giang NH, and Tuyen ND. "From Offshore Wind Pipeline to Bankable Power: Managing Grid, Curtailment and Offtake Risk in ASEAN." Book chapter, *ASEAN Centre for Energy*, 2026. (Accepted, in preparation)
2. Long PV, **Long HH**, and Tuyen ND. "A Comprehensive Review of Direct Coupled Photovoltaic–Electrolyser System: Sizing Techniques, Operating Strategies, Research Progress, Current Challenges, and Future Recommendations." *International Journal of Hydrogen Energy* (Q1), 48, 25231–25249, 2023.
3. Long PV, **Long HH**, Kien DC, Hirotaka T, and Tuyen ND. "An Improved State Machine-Based Energy Management Strategy for Renewable Energy Microgrid with Hydrogen Storage System." *Energy Reports* (Q1), 9 (Suppl. 1), 194–201, 2023.
4. **Long HH**, Long PV, and Nguyen D. "Power to Hydrogen Life Cycle Assessment: Method and a Case Study for the United States." *WindEurope Annual Event*, Copenhagen, 2025.
5. **Long HH**, Anh NH, Hirotaka T, Goro F, and Tuyen ND. "A Small Scale Microgrid Experimental Model Integrated Battery–Hydrogen Storage System." *IEEJ* Conference, Japan, 2024.
6. Kien DC, Anh NH, **Long HH**, Long PV, Hirotaka T, and Tuyen ND. "Grid Forming-Based Hydrogen System for High-Penetration Renewable Energy Microgrid." *International Conference on Power and Energy Systems Engineering (CPESE)*, Japan, 2024.

Skills and Tools

Power-system analysis:	Reactive power and loss studies, microgrid control – *PowerFactory, MATLAB/Simulink*
Electrical design:	Cable systems, substation concepts, BESS sizing – *QGIS, AutoCAD, Visio*
Techno-economics:	CAPEX, NPV option studies, due diligence – *Excel/VBA cost models*
Programming:	Optimisation and automation – *Python, VBA, MATLAB, LaTeX*
Languages:	English (fluent), Vietnamese (native)

Honours and Activities

OWC Scholarship, top 5 of 200 students in the School 2024
Third Prize, Student Forum, top 30 of 1,000+ student research papers in Vietnam 2022
Delegate, Youth Engagement in Just Energy Transition, UNDP Vietnam 2023
Secretary, HUST Youth Union 2020–2024
Third Prize, Vietnam Mathematical Olympiad, top 45 of 2,000+ in the national round 2020